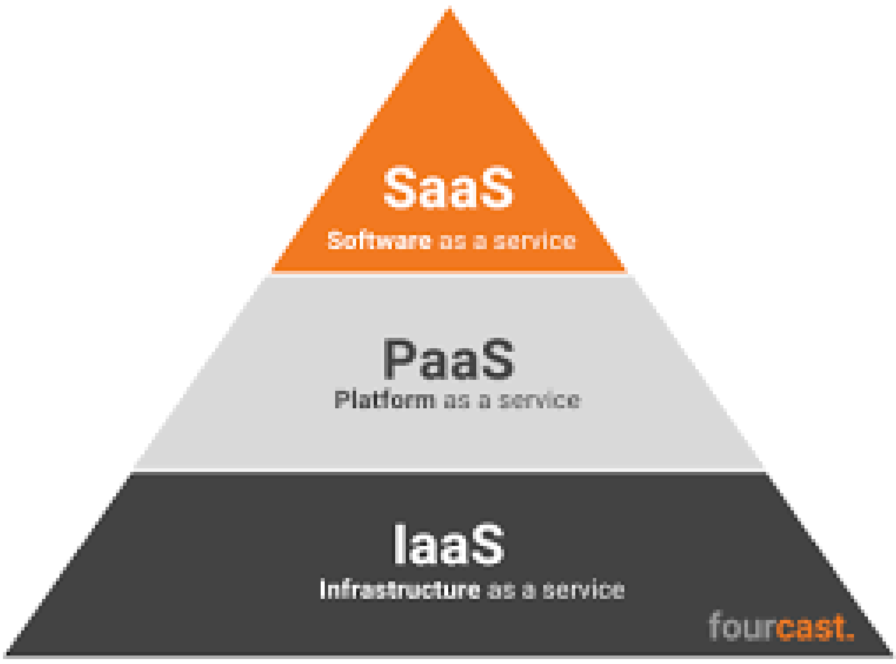
# *CLOUD COMPUTING SERVICE MODELS*

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# *IaaS, PaaS and SaaS*

Infrastructure as a service (IaaS), platform as a service (PaaS) and software as a service (SaaS) are the three most popular types of cloud service offerings. They are sometimes referred to as the three core cloud service models or cloud computing service models.

"As a service" refers to the way IT assets are consumed with cloud-based products, highlighting the essential difference between cloud computing and traditional IT. In traditional IT, businesses consume IT assets by purchasing, installing, managing and maintaining them in on-premises data centers. With cloud computing, the cloud service provider owns, manages and maintains the assets.

IaaS is a form of cloud computing that delivers on-demand access to cloud-hosted compute, storage and networking—the backend IT infrastructure for running applications and workloads in the cloud. It enables businesses to scale resources as needed, and reduces the need for significant upfront capital expenditures or complex on-premises infrastructure configurations.

Businesses often rely on IaaS tools to manage their high-performance workloads, especially in the case of "spiky" workloads that are prone to sudden surges of user activity.

PaaS is a cloud computing model that provides a complete on-demand cloud platform (hardware, software and infrastructure) for developing, running, maintaining and managing applications. The PaaS provider hosts everything—including servers, networks, data storage, operating system (OS) software, databases and development tools—at their data center.

PaaS enables businesses to build, test, deploy, run, update and scale applications faster and cheaper than they would with an internally developed and managed, on-premises platform.

SaaS is cloud-hosted application software delivered over the internet to compatible computing devices. SaaS providers operate, manage and maintain the software and the infrastructure on which it runs. Instead of buying application software and installing on a local device, SaaS users simply create an account, subscribe to the application and get to work.

IaaS, PaaS and SaaS are not mutually exclusive—many enterprises use all three—as each one provides developers accessible, scalable IT capabilities with a more predictable cost structure

# Infrastructure as a service (IaaS)

IaaS offers a cloud-based alternative to on-premises computing infrastructure, delivering physical and virtual computing resources (hosted in data centers by IaaS providers) to users.

With IaaS, providers supply the foundation for IT operations (the infrastructure, including maintenance, patches, upgrades and troubleshooting), so customers can focus on software and strategy rather than managing physical hardware. Customers access IT resources and services over a wide area network—typically the internet—and use the cloud provider's platform to manage the rest of the application stack.

IaaS tools rely on virtualization technology, which enables multiple virtual machines (VMs) to run on a single physical server. Each VM operates independently—with its own operating system, applications, and memory and storage specifications—and the provider manages hypervisors (also called virtual machine monitors or VMMs) that keep the VMs isolated. Hypervisors also allocate computing power, memory and storage to each instance, enabling customers to create, configure and scale virtual instances to match their needs.

Depending on business needs, IaaS can be paired with automated services such as auto scaling, load balancing, disaster recovery and performance monitoring to help optimize application availability and workload management. IaaS platforms can also help businesses take advantage of AI and machine learning (ML) technologies. Developers can, for example, use the computing power offered through IaaS to create foundation models for building and scaling generative AI applications.

Flexible, usage-based (pay-as-you-go) models are the most common for IaaS. But for more predictable workloads and longer-term commitments, companies can opt for reserved instances or subscription-based billing (such as monthly billing). Many providers offer discounts for these commitments.

## Types of IaaS resources

IaaS delivers a range of virtualized computing resources that users can access and control over the internet. They include:

- Compute resources
- Bare metal servers
- Virtual servers
- Networking
- Storage
- Containers
- Security

## IaaS use cases

- **Backup and disaster recovery**

IaaS provides cloud-based backup and disaster recovery solutions, enabling organizations to replicate and back up their systems and data in the cloud to help ensure business continuity.

- **Big data analytics**

IaaS platforms can provide the enormous processing power businesses need to analyze big data and make data-driven decisions.

- **Website hosting**

IaaS offers businesses a cost-efficient way to host secure, scalable customer-facing websites and applications, and to deliver fast, consistent end user experiences.

- **High-performance computing (HPC)**

Compared to a traditional on-premises infrastructure setup, IaaS provides business an efficient and cost-effective way to support HPC, a technology that uses clusters of powerful processors that work in parallel to process massive, multidimensional datasets and solve complex problems at ultra-fast speeds.

- **Hybrid cloud and multicloud migration**

IaaS can facilitate the deployment of resources across hybrid cloud environments. For instance, IaaS supports "lift and shift" migration, where workloads are moved from an on-premises setting to a cloud provider's data center.

# Platform as a service (PaaS)

PaaS provides a cloud-based platform for developing, running, managing applications. The cloud services provider hosts, manages and maintains all the hardware and software included in the platform—servers (for development, testing and deployment), operating system (OS) software, storage, networking, databases, middleware, runtimes, frameworks and development tools—as well as related services for security, operating system and software upgrades, backups and more.

Users access the PaaS through a graphical user interface (GUI), where development or DevOps teams can collaborate on all their work across the entire application lifecycle including coding, integration, testing, delivery, deployment and feedback.

With PaaS, the customer can pay a fixed fee for a set amount of resources and a specific number of users, or like IaaS customers, they can choose pay-as-you-go pricing and pay only for the resources they use.

## PaaS components

**Cloud infrastructure**

Cloud infrastructure is the backbone of any PaaS system. It includes VMs, OSs, storage and security features, such as firewalls and encryption.

**Application management tools**

PaaS provides tools for creating, launching and managing software products. In PaaS environments, applications are often created using middleware that streamlines workflows and enables multiple development and operations teams to work on the same project at once.

**Graphical user interfaces (GUIs)**

GUIs are user-friendly dashboards where development or DevOps teams can manage their work throughout the application lifecycle. They serve as the main point of interaction between developers and a PaaS environment.

With PaaS, all standard development tools are available through a GUI, enabling developers to log in from anywhere, collaborate on projects, test new applications and release finished products.

## PaaS use cases

- **API development and management**

Because of its built-in frameworks, PaaS can simplify API development, deployment and management, enabling data and feature sharing between applications.

- **Agile development and DevOps**

PaaS provides fully configured environments that facilitate process automation for testing, security and deployment throughout the software application lifecycle. These features help support CI/CD pipelines and agile development practices.

- **Cloud migration and cloud-native development**

With its ready-to-use tools and integration capabilities, PaaS can simplify the migration of existing applications to the cloud. Specifically, PaaS supports re-platforming—moving an application to the cloud with modifications that leverage cloud scalability, load balancing and other capabilities.

PaaS also supports refactoring, or the rearchitecting of part or all of an application, through technologies that facilitate cloud-native development (such as microservices, containers and serverless functions).

- **Internet of Things applications**

PaaS solutions can support a range of programming languages (Java and Python, for example), tools and application environments used for IoT app development and real-time IoT data processing.

- **Hybrid cloud strategy**

PaaS solutions can enable developers to build apps one time, then deploy and manage them anywhere in their hybrid environment.

- **Enterprise AI models**

A sustainable, fully provisioned, distributed architecture helps improve and future-proof the performance of enterprise-grade AI tools, including generative AI. PaaS provides such an environment, supporting and streamlining the development and deployment of AI applications.

# Software as a service (SaaS)

Sometimes called cloud application services, SaaS offers complete, provider-managed software solutions delivered over the internet.

SaaS leverages cloud computing and economies of scale to offer customers a more efficient way to access, use and pay for software. Cloud providers host SaaS applications on their servers and manage the availability, security and overall performance of applications, including any software updates or feature enhancements. Users typically subscribe to SaaS applications on a monthly or annual basis and access them using a web browser or mobile apps.

Some SaaS providers offer usage-based pricing, but in many cases, customers can choose a flat-rate pricing model. Flat-rate pricing is a “one plan, one price” approach, where users can access all app features for a single, fixed price, like a streaming service subscription. The user pays a monthly or annual fee for unlimited use, regardless of how much they use the application. Many SaaS providers offer tiered packages based on the number of users that will need access to the app.

## SaaS features

**Cloud-based architecture**
SaaS applications are designed to run in the cloud. Vendors can host these applications on their own infrastructure or use established cloud providers, but relying on established providers enables SaaS developers to maximize app scalability and extend the app’s reach to a broader customer base.

**Universal accessibility**
SaaS applications are available to any user with an internet connection and a compatible device—whether it's a computer, smartphone or tablet. Most SaaS solutions run in web browsers, though some require users to install dedicated mobile apps or lightweight clients on their devices.

**Multitenant model**
SaaS apps run on a multitenant architecture, where a single instance of the application serves multiple customers. To help ensure security and privacy, each customer’s data and configurations are kept separate from the others.

**Minimal customer maintenance**

With SaaS, customers don't have to worry about managing or maintaining the underlying infrastructure because the cloud provider delivers all the critical services.

**Integration tools**

Many providers offer APIs that enable customers to integrate SaaS solutions with other cloud-based web applications or on-premises software systems (connecting an e-commerce site with a "buy now, pay later" platform, for instance).

## SaaS - Use Cases

Much of the software that today's workforces use is delivered through the SaaS model, including:

- **Everyday tools**, such as Slack (for messaging) and Dropbox (for file storage and sharing).

- **Core business applications**, such as enterprise resource planning (ERP), human resources management systems (HRMSs) and customer relationship management applications.
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- **Workforce optimization platforms**, such as Workday (for financial management and workforce planning) and Jira (for project management and workflow automation).

## SaaS benefits

SaaS applications also come with their own lock-in challenges. Proprietary technologies, complex integrations (using vendor-specific APIs, for example) and other factors can create application dependencies that make it complicated and expensive to change SaaS providers.

Risks aside, SaaS platforms can enable:

- **Faster adoption.** Customers can purchase and start using SaaS applications immediately, sometimes in minutes, for a minimal upfront cost (typically, the cost of the first month's subscription).

- **Faster upgrades and updates.** SaaS applications give users access to new features and versions as soon as they're available. Providers often upgrade features and add functions several times weekly, often without customers noticing.

- **On-demand scalability.** Users can scale SaaS applications up and down as needed, by simply upgrading or downgrading tiers or purchasing more capacity.